



Netrokona University
Netrokona 2400, Bangladesh
Department of Computer Science and Engineering
Faculty of Engineering and Technology
CSE-3112: Software Engineering Lab

Requirements Gathering: Public Voice - Smart Citizen
Complaint Management System

Team 2

Naimul Islam Maruf (202204016)

Hridoy Kumar Pramanik (202204022)

Nazmus Sakib (202204028)

Session: 2022-23 (3rd Year 1st Semester)

Requirements Gathering

Public Voice - Smart Citizen Complaint Management System

1. Introduction

Purpose

The purpose of this system is to provide a centralized, secure, and transparent platform where citizens can submit complaints related to public services, safety, consumer rights, health, environmental issues, and other civic matters. The system enables government authorities to receive, investigate, manage, and resolve complaints efficiently while maintaining accountability through automated escalation and progress tracking.

Objectives

- Allow citizens to submit complaints online.
- Reduce manual paperwork.
- Ensure every complaint reaches the appropriate authority.
- Automatically escalate unresolved complaints.
- Improve transparency.
- Reduce corruption and negligence.
- Maintain digital records.
- Enable government officials to monitor complaint resolution.

2. Stakeholders

Primary Stakeholders

- Citizens
- Local Police
- District Police
- District Administration
- Ministry Officials
- Government Departments
- System Administrator

Secondary Stakeholders

- Investigation Officers
- Food Safety Authority
- Drug Administration
- Consumer Rights Authority
- Environmental Authority

3. User Roles

Citizen: A registered citizen who submits complaints.

- Register
- Login
- Submit complaint
- Upload evidence
- Track complaint
- Edit profile
- View complaint history
- Receive notifications
- Give feedback
- Appeal rejected complaints

Local Officer: Responsible for investigating complaints within a local jurisdiction.

- Login
- View assigned complaints
- Accept complaint
- Reject complaint with reason
- Request additional evidence
- Update complaint status
- Assign field officers
- Upload investigation report

District Officer (SP/DC)

- Monitor all complaints within district
- Reassign complaints
- View reports
- Escalate complaints
- Monitor officer performance

Ministry Officer

- View nationwide complaints
- Generate reports
- Analyze statistics
- Monitor district performance

System Administrator

- Manage users
- Manage officers
- Manage complaint categories
- Configure escalation rules
- Manage districts
- Manage departments
- View complete logs
- Backup database

4. Functional Requirements

User Registration: The system shall allow citizens to register using-

- Full Name
- NID Number
- Date of Birth
- Mobile Number
- Email
- Password
- Address
- District
- Division

Authentication: The system shall-

- Allow secure login
- Encrypt passwords
- Support role-based login
- Logout securely
- Reset password

Complaint Submission: The system shall allow users to submit-

- Complaint Title
- Complaint Description
- Category
- Department
- Location
- Date of Incident
- Priority
- Evidence Upload
- GPS Location

Complaint Categories: The system shall support

- Food Safety
- Expired Products
- Medicine Complaint
- Medical Negligence
- Road Damage
- Corruption
- Cyber Crime
- Domestic Violence
- Harassment
- Electricity
- Water Supply
- Environmental Pollution
- Illegal Construction
- Garbage
- Consumer Rights
- Noise Pollution
- Public Transport
- Education
- Others

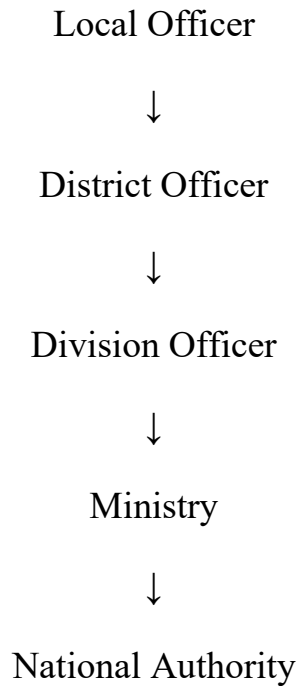
Evidence Upload

- Images
- PDF
- Videos
- Audio
- Documents

Complaint Tracking

- Users shall be able to track
- Submitted
- Accepted
- Assigned
- Under Investigation
- Evidence Verified
- Resolved
- Closed
- Rejected
- Escalated

Escalation System: If complaint remains unresolved



Every escalation shall be logged.

Notification System

- Users shall receive notifications
- Complaint submitted
- Complaint accepted
- Complaint rejected
- Complaint updated
- Complaint resolved
- Complaint escalated

Complaint History

- Citizen shall view
- Current complaints
- Resolved complaints
- Rejected complaints
- Archived complaints

Search Functionality

- Users shall search by
- Complaint ID
- Category
- Date
- Location
- Status
- Department
- Officer
- District
- Citizen Name (Admin only)

Feedback System

- Citizen can
- Rate officer
- Write review
- Report dissatisfaction

Dashboard

Citizen Dashboard

- Total Complaints
- Pending
- Solved
- Rejected
- Notifications

Officer Dashboard

- Assigned Cases
- Pending Cases
- Today's Cases
- Urgent Cases

Report Generation

- System shall generate
- Daily Reports
- Monthly Reports
- Yearly Reports
- Department Reports
- District Reports
- Officer Performance Reports
- Complaint Statistics

Admin Dashboard

- Nationwide Statistics
- Complaint Trends
- District Rankings
- Department Performance

Audit Log

- Every important activity shall be recorded.
- Examples
- User Login
- Complaint Submission
- Complaint Update
- Complaint Escalation
- Officer Assignment
- Admin Actions

5. Non-Functional Requirements

Performance

- Response time below 3 seconds
- Support at least 1000 concurrent users
- Fast complaint search

Reliability

- 99% uptime
- Automatic backup
- Error recovery

Availability: 24/7 access

Security

- Password hashing
- Role-based access
- Input validation
- SQL Injection protection
- XSS protection
- CSRF protection
- Secure sessions
- HTTPS support

Scalability

- Support millions of complaints
- Support additional departments
- Support additional districts

Maintainability

- Modular architecture
- Well documented
- Easy bug fixing
- Easy feature addition

Usability

- Simple interface
- Responsive design
- Mobile friendly
- Easy navigation
- Minimal learning curve

Compatibility

- Chrome
- Firefox
- Edge
- Safari
- Android
- iOS
- Windows
- Linux

6. Hardware Requirements

- Server
- CPU
- Minimum Quad Core
- RAM
- 8 GB minimum
- Storage
- 250 GB SSD
- Internet
- Stable broadband

Client

- Laptop/Desktop/Smartphone
- Browser
- Internet connection

7. Software Requirements

Frontend

- HTML5
- CSS3
- Bootstrap
- JavaScript
- React.js (optional)

Backend

- Node.js / PHP
- Express.js (if Node)

Database

- MySQL / MongoDB
- Server, Xampp
- Apache

IDE

- VS Code
- Version Control
- Git
- GitHub

8. Business Rules

- One NID can register only one account.
- Every complaint receives a unique Complaint ID.
- Complaints cannot be permanently deleted by users.
- Every complaint must belong to exactly one category.
- Every complaint must be assigned to one responsible department.
- Every complaint status change is logged.
- Users cannot edit complaints after investigation begins.
- Officers cannot investigate complaints outside their jurisdiction unless reassigned.
- Rejected complaints must include a written reason.
- All uploaded evidence is stored securely and linked to the complaint.

9. Assumptions

- Citizens provide valid information.
- Internet connectivity is available.
- Government officers regularly access the system.
- Digital records are accepted for processing.
- Simulated NID verification is sufficient for the academic version.

10. Constraints

- Real government NID APIs are not accessible in the student version.
- SMS and email notifications may be simulated.
- Actual integration with police or ministry systems is outside the project scope.
- Payment gateways and legal enforcement are not implemented.

Bonus Features for Future Improvements

- **Complaint ID Tracker** similar to courier tracking.
- **Complaint Heat Map** showing complaint density by area.
- **Officer Performance Dashboard** with resolution metrics.
- **Automatic escalation** after a configurable number of days.
- **Duplicate complaint detection** to reduce repeated reports.
- **Anonymous reporting** for sensitive categories (while still verifying the user internally).
- **Evidence verification workflow** with approval/rejection by officers.
- **Citizen satisfaction rating** after complaint resolution.